Routing, capacity and width in dynamic convolution: ResNet-50

Replication of the 132-run ResNet-18 program

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Generated from result files by `report/build_model_report.py`

SYNTHETIC TEST DATA. Every number in this document is generated to test the pipeline. None is a result.

This document reports one backbone of the cross-architecture replication. It is generated from the result files: every number, table and figure is computed, and every sentence states what a test returned. Interpretation across backbones is in the cross-model report. Method, statistical protocol and the pre-registered margin are those of the ResNet-18 study and are not repeated here.

1 Setup and coverage

Backbone. ResNet-50, adapted to 32×32 and 64×64 inputs as recorded in `src/models`. Width 64 is the standard network; widths 32 and 16 scale every channel count by $1/2$ and $1/4$. Dynamic layers follow the ResNet-18 rule literally (every $k \times k$ convolution with $k > 1$ inside the blocks; stem, 1×1 convolutions, shortcuts, downsampling layers and head static): here the 3×3 convolution of every bottleneck block, 16 layers. Initialisation: Kaiming-normal (fan-out) for every static convolution and every bank kernel.

Size. At CIFAR-100 width 64 the static network stores 23.71 M parameters and the $K = 4$ `gap` network 57.98 M, of which 319.2 K are router parameters (Table 14). The bank adds 33.95 M to the static network; the router is 1.3 % of the static network's size. The router is the ResNet-18 design (a two-layer MLP with hidden width $C_{\text{in}}/4$ on the global descriptor), so its size grows with the square of a layer's input channels while a depthwise kernel grows linearly: where dynamic layers are depthwise, the routed arms carry many more parameters than the bank. The `gap−uniform` routing term includes them.

Gate. The first run, static, CIFAR-100, width 64, seed 0, reached 79.00 % against a pre-registered floor of 70.0 %: passed.

Coverage. 130 of 132 runs are present; 2 are missing, of which 1 diverged (non-finite training loss). 0 group(s) ran with gradient accumulation because batch 128 did not fit in memory. Table 1 lists every missing run and every deviation. Paired tests use the seeds both arms have; a verdict computed on fewer than six pairs carries its n.

Table 1: Missing runs, divergences and deviations from the recipe.

dataset	width	K	τ	arm	seed	status
CIFAR-100	32	4	anneal	`gap`	3	not run
Tiny-ImageNet	32	4	const	`gap`	0	diverged (non-finite loss)

2 CIFAR-100 at the standard width

Against `static`: `uniform` +0.007 pp (equivalent); `gap` +0.744 pp (higher); `shuffled` +0.094 pp (equivalent); `frozen_random` +0.039 pp (equivalent). 3 of the four arms are equivalent to `static` within 0.5 pp. The largest gap between arm means is 0.744 pp against a median within-arm seed standard deviation of 0.160 pp; one-way ANOVA across the five arms $F = 19.150$, $p = 0.000$. `frozen_random−gap` $= -0.706$ pp ($p = 0.001$, $p_{\text{TOST}} = 0.9413$). Power for `gap−uniform` here: paired sd 0.194 pp, minimum detectable difference 0.204 pp, simulated power 71.7, 99.8 and 100.0 % at true effects of 0.25, 0.5 and 1.0 pp. Table 2 gives the routing entropy by arm.

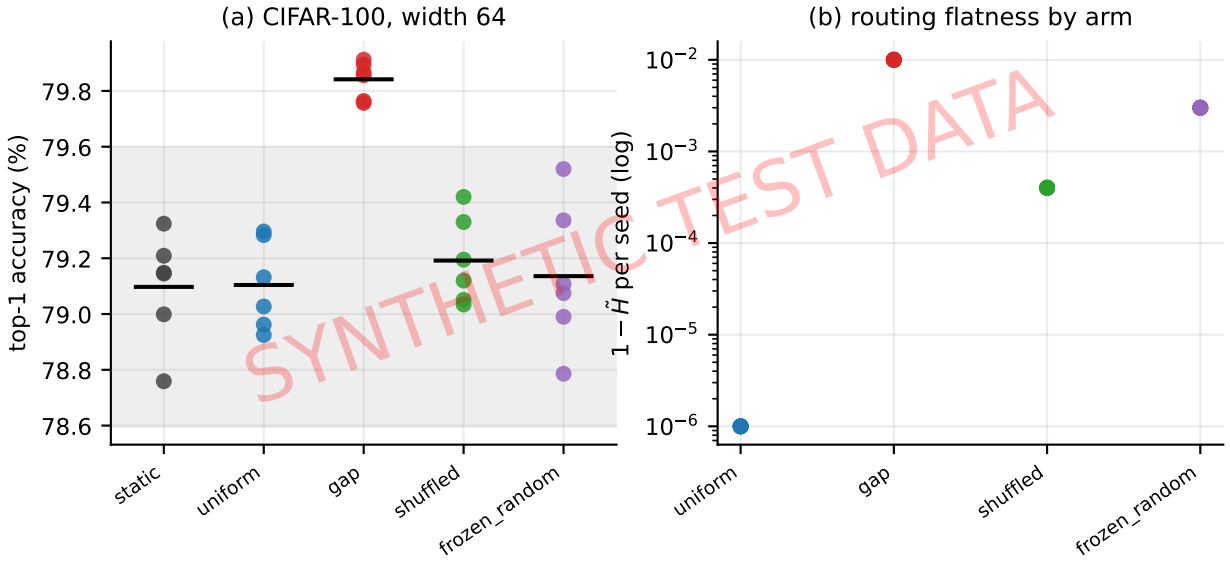


Figure 1: CIFAR-100, width 64. (a) Every run of the five arms; bars are arm means and the shaded band is ±0.5 pp around `static`. (b) Router flatness $1 - \tilde{H}$ per seed, log scale; `uniform` is exactly zero and is drawn at the axis floor.

Table 2: Routing entropy by arm, CIFAR-100 width 64, per-seed means.

arm	mean $\tilde{H}$	min	max	$1 - \tilde{H}$
`uniform`	1.0000	1.0000	1.0000	0.00000
`gap`	0.9900	0.9900	0.9900	0.01000
`shuffled`	0.9996	0.9996	0.9996	0.00040
`frozen_random`	0.9970	0.9970	0.9970	0.00300

3 The capacity sweep on CIFAR-100

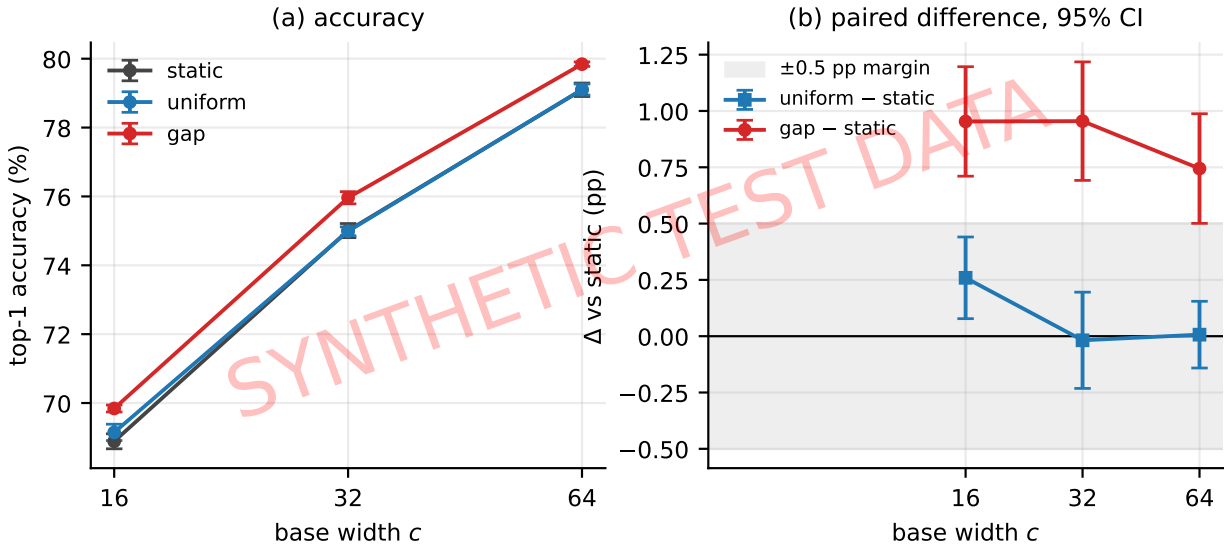


Figure 2: CIFAR-100. (a) Accuracy against base width, ±1 standard deviation over six seeds. (b) Paired difference against `static` with 95 % confidence intervals; the band is the ±0.5 pp margin.

`gap−static`: +0.954 pp at width 16 (higher), +0.955 pp at width 32 (higher), +0.744 pp at width 64 (higher) (Figure 2, Table 13).

4 Bank against routing

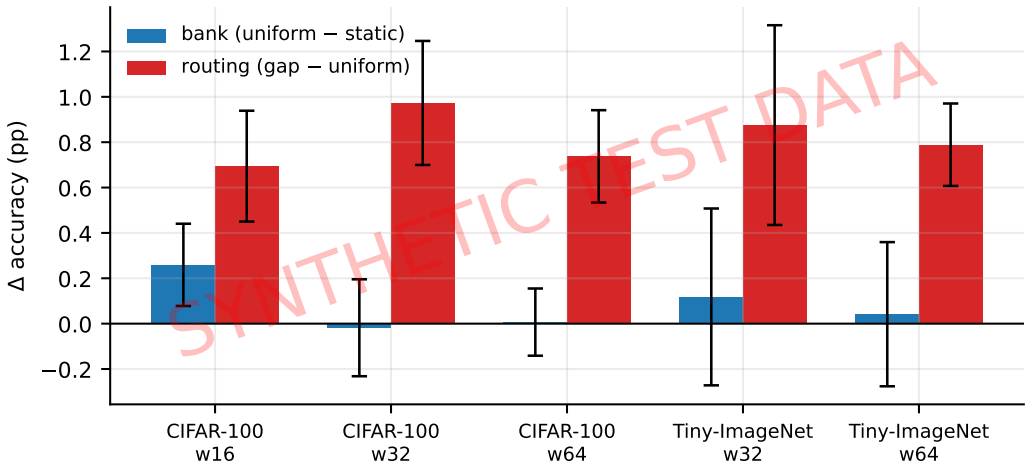


Figure 3: The total effect split into a bank term (`uniform−static`) and a routing term (`gap−uniform`), five conditions, 95 % confidence intervals.

The bank term is equivalent to zero in 4 of 5 conditions; the others: CIFAR-100 width 16: higher (+0.259 pp). The routing term is positive with $p < 0.05$ in 5 of 5 conditions: CIFAR-100 width 16 +0.694 pp ($p = 0.0008$); CIFAR-100 width 32 +0.973 pp ($p = 0.0003$); CIFAR-100 width 64 +0.737 pp ($p = 0.0002$); Tiny-ImageNet width 32 +0.875 pp ($p = 0.0053$); Tiny-ImageNet width 64 +0.789 pp ($p = 0.0001$) (Table 3, Figure 3).

Table 3: Decomposition. Bank verdicts use TOST at ±0.5 pp.

		bank (`uniform−static`)			routing (`gap−uniform`)		total	
dataset	width	Δ (pp)	p_{TOST}	verdict	Δ (pp)	p	Δ (pp)	p
CIFAR-100	16	+0.259	0.0095	higher	+0.694	0.0008	+0.954	0.0002
CIFAR-100	32	-0.018	0.0011	equivalent	+0.973	0.0003	+0.955	0.0002
CIFAR-100	64	+0.007	0.0002	equivalent	+0.737	0.0002	+0.744	0.0005
Tiny-ImageNet	32	+0.118	0.0266	equivalent	+0.875	0.0053	+1.044	0.0002
Tiny-ImageNet	64	+0.042	0.0069	equivalent	+0.789	0.0001	+0.830	0.0021

5 Bank size

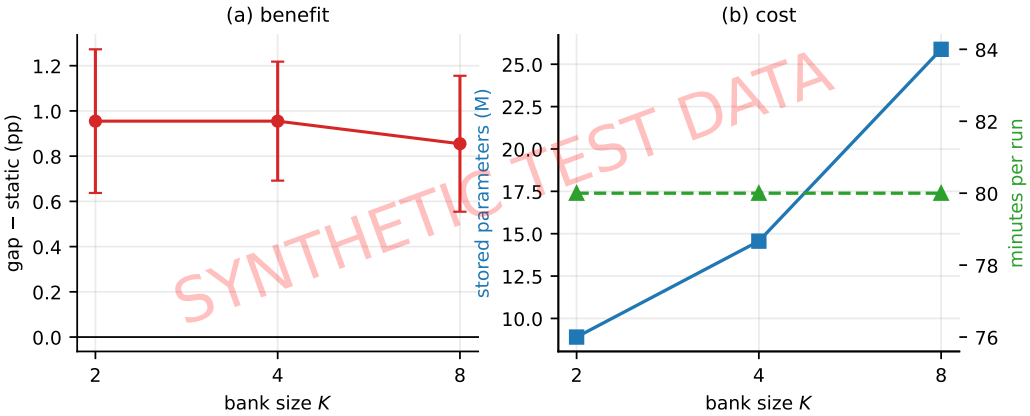


Figure 4: CIFAR-100 width 32, `gap` with $K = 2, 4, 8$. (a) Difference against `static`, 95 % CI. (b) Stored parameters and minutes per run.

$K = 8$ against $K = 2$: -0.100 pp (equivalent) for 2.91× the parameters (Tables 4, 5).

Table 4: Bank size at CIFAR-100 width 32. Differences are against `static`.

K	accuracy	sd	Δ (pp)	95 % CI	p	params (M)	min/run
2	75.962	0.213	+0.955	[+0.637, +1.272]	0.0006	8.90	80.0
4	75.962	0.176	+0.955	[+0.692, +1.218]	0.0002	14.56	80.0
8	75.862	0.174	+0.855	[+0.554, +1.155]	0.0008	25.88	80.0
`static`	75.007	0.205	–	–	–	5.99	20.0

Table 5: Pairwise comparisons between bank sizes.

comparison	Δ (pp)	95 % CI	p	p_{TOST}	verdict
$K = 4$ vs $K = 2$	+0.000	[-0.261, +0.261]	1.0000	0.0022	equivalent
$K = 8$ vs $K = 4$	-0.100	[-0.313, +0.113]	0.2812	0.0024	equivalent
$K = 8$ vs $K = 2$	-0.100	[-0.442, +0.242]	0.4858	0.0149	equivalent

6 Tiny-ImageNet

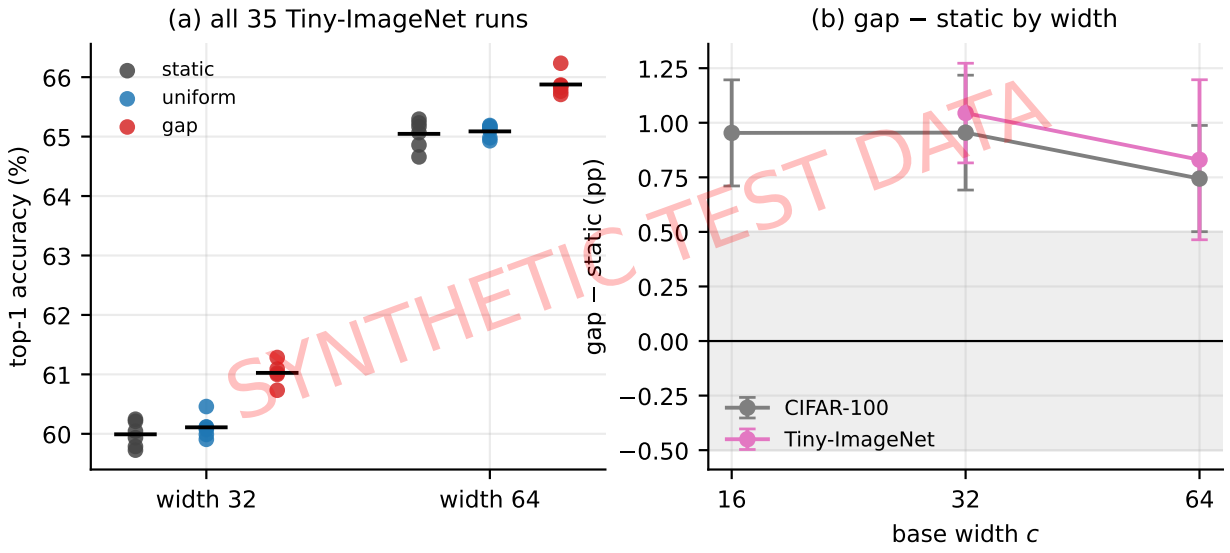


Figure 5: Tiny-ImageNet. (a) Every run at widths 32 and 64. (b) `gap−static` against width on both datasets, 95 % CI; the band is the margin.

`gap−static` is +1.044 pp at width 32 (higher (n=5)) and +0.830 pp at width 64 (higher).

Across the five conditions the Spearman correlation between the static arm's final training loss and `gap−static` is $\rho = +0.70$ (Table 6, Figure 6).

Table 6: Five conditions ordered by the final training loss of the static arm.

dataset	width	static train loss	`gap − static` (pp)	95 % CI	verdict
CIFAR-100	64	0.0244	+0.744	[+0.501, +0.988]	higher
Tiny-ImageNet	64	0.0306	+0.830	[+0.464, +1.197]	higher
CIFAR-100	32	0.0619	+0.955	[+0.692, +1.218]	higher
Tiny-ImageNet	32	0.0869	+1.044	[+0.816, +1.273]	higher (n=5)
CIFAR-100	16	0.1119	+0.954	[+0.710, +1.197]	higher

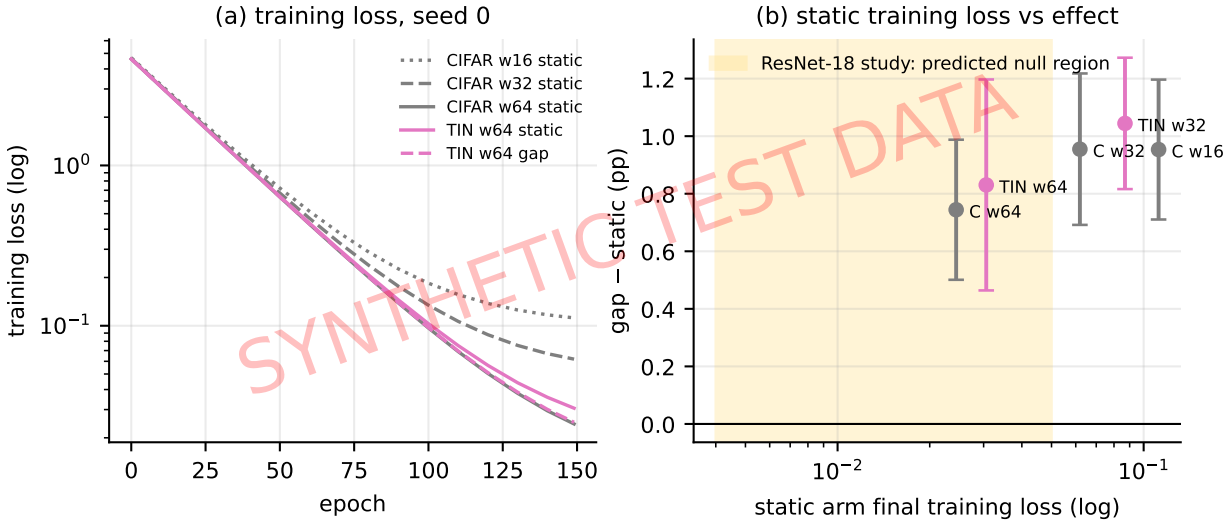


Figure 6: (a) Training loss, seed 0, sampled every ten epochs. (b) `gap−static` against the static arm's final training loss; the shaded region is where the ResNet-18 study predicted a null.

7 Equal parameter budgets

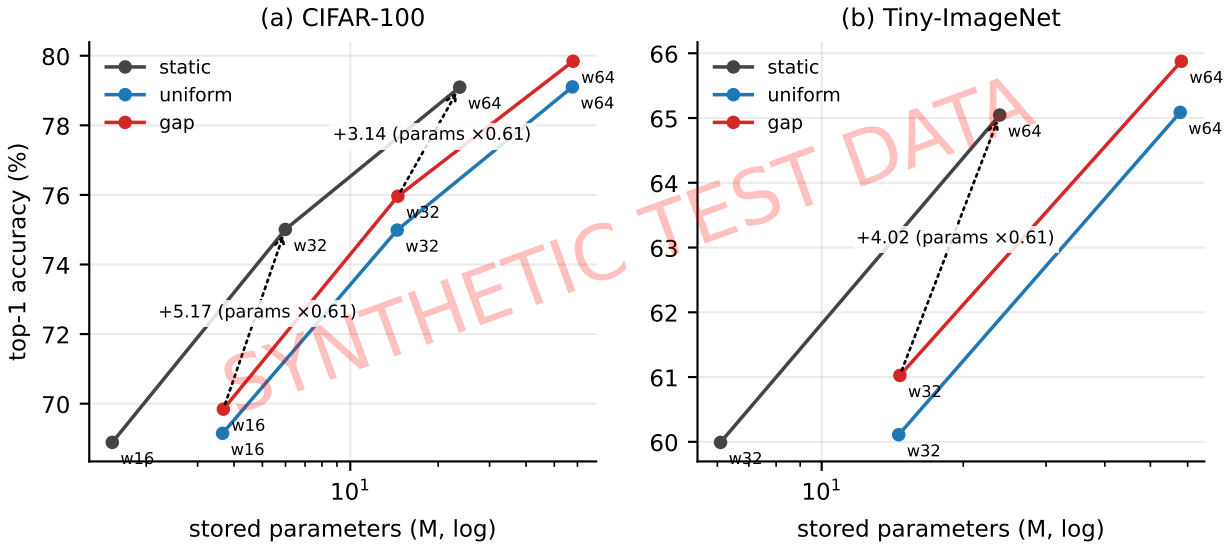


Figure 7: Accuracy against stored parameters. Arrows join `gap` at width c to `static` at width $2c$; the label is the accuracy difference, and the parameter ratio where it is outside $\pm 10\,\%$.

The ResNet-18 comparison of `gap` at width c with `static` at width $2c$ was a matched-budget comparison because a $K = 4$ bank and a doubling of width both multiply the parameters of the dynamic layers by four. That holds only where almost every parameter is in a dynamic layer. Here 0 of 3 pairs are within 10 % and are called matched: CIFAR-100 `gap` w16 vs `static` w32: +5.166 pp, parameter ratio 0.61 (not matched); CIFAR-100 `gap` w32 vs `static` w64: +3.135 pp, parameter ratio 0.61 (not matched); Tiny-ImageNet `gap` w32 vs `static` w64: +4.017 pp, parameter ratio 0.61 (not matched) (Table 7). Table 8 adds the comparison at equal parameters: the static accuracy curve interpolated linearly in log-parameters between the static widths that were run, evaluated at the `gap` network's count. It is an interpolation, not a run, and has no confidence interval.

Table 7: `gap` at width c against `static` at width $2c$. Δ is plain minus routed. The last column is the routed network's parameters divided by the plain one's; a pair is called matched only within 10 %.

dataset	routed	plain	routed	plain	Δ (pp)	95 % CI	p	min/run	param ratio
CIFAR-100	`gap` w16 (3.68M)	`static` w32 (5.99M)	69.841	75.007	+5.166	[+4.986, +5.347]	$< 10^{-4}$	40.0 / 20.0	0.61 (not matched)
CIFAR-100	`gap` w32 (14.56M)	`static` w64 (23.71M)	75.962	79.097	+3.135	[+2.844, +3.426]	$< 10^{-4}$	80.0 / 40.0	0.61 (not matched)
Tiny-ImageNet	`gap` w32 (14.66M)	`static` w64 (23.91M)	61.025	65.046	+4.017	[+3.793, +4.240]	$< 10^{-4}$	80.0 / 40.0	0.61 (not matched)

Table 8: `gap` against the static accuracy curve at equal parameters (interpolated, no CI).

dataset	routed	params (M)	routed acc.	static at equal params	Δ (pp)
CIFAR-100	`gap` w16	3.68	69.841	72.816	+2.975
CIFAR-100	`gap` w32	14.56	75.962	77.648	+1.686
Tiny-ImageNet	`gap` w32	14.66	61.025	63.238	+2.213

8 Temperature schedule

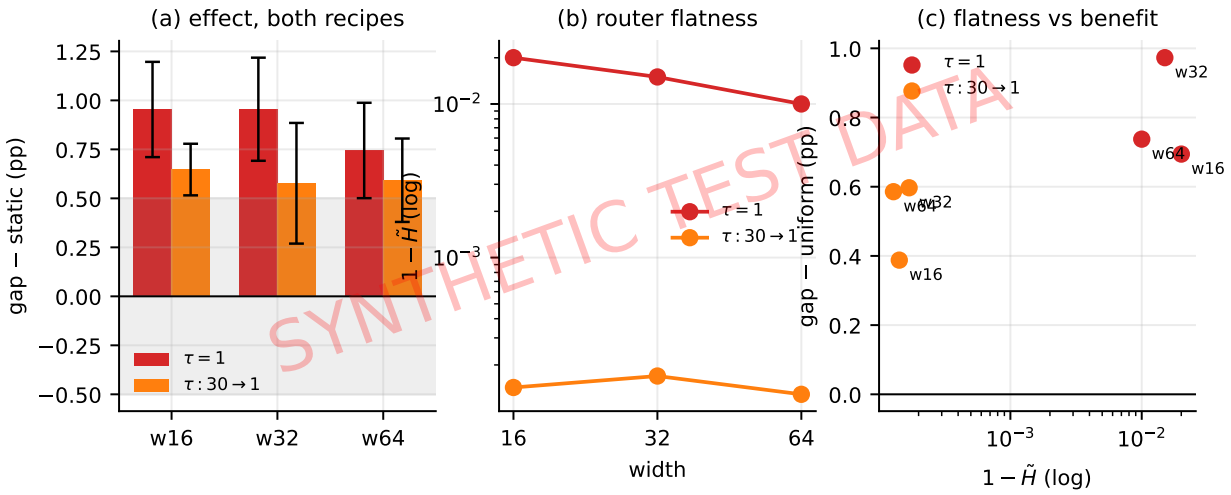


Figure 8: Constant $\tau = 1$ against τ annealed from 30 to 1 over ten epochs, CIFAR-100. (a) `gap−static`. (b) Router flatness. (c) Flatness against benefit over `uniform`.

Annealed minus constant, `gap`: width 16 -0.307 pp (lower); width 32 -0.358 pp (inconclusive (n=5)); width 64 -0.152 pp (equivalent) (Table 9).

Table 9: Temperature schedule, CIFAR-100.

	accuracy		annealed − constant		vs `static`			annealed vs `uniform`	
width	$\tau = 1$	annealed	Δ	p	$\tau = 1$	annealed	p_{TOST}	Δ	verdict
16	69.841	69.535	-0.307	0.0448	+0.954	+0.647	0.9825	+0.388	higher
32	75.962	75.619	-0.358	0.0585	+0.955	+0.577	0.7372	+0.597	higher (n=5)
64	79.842	79.690	-0.152	0.0711	+0.744	+0.593	0.8427	+0.586	higher

9 Routing entropy and routing benefit

At width 16 the constant-temperature router's $1 - \tilde{H}$ is 140.4 times the annealed router's; the benefit over `uniform` is +0.694 pp constant and +0.388 pp annealed. Across the 6 width–recipe points the Spearman correlation between $1 - \tilde{H}$ and `gap−uniform` is $\rho = +0.77$ (Table 10).

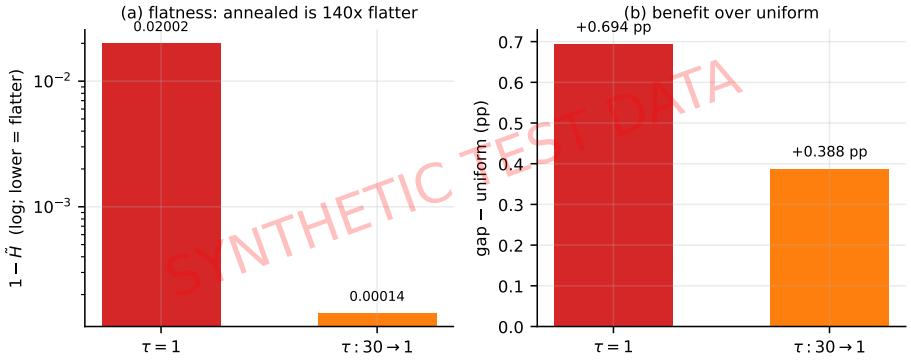


Figure 9: CIFAR-100 width 16, both temperature recipes. (a) Router flatness. (b) Benefit of `gap` over `uniform`.

Table 10: Routing entropy against the benefit over `uniform`.

width	τ	$\tilde{H}$	$1-\tilde{H}$	RMS dev	`gap−uniform` (pp)	p
16	1	0.97998	0.02002	0.0589	+0.694	0.0008
16	$30 \to 1$	0.99986	0.00014	0.0050	+0.388	0.0051
32	1	0.98500	0.01500	0.0510	+0.973	0.0003
32	$30 \to 1$	0.99983	0.00017	0.0054	+0.597	0.0011
64	1	0.99000	0.01000	0.0416	+0.737	0.0002
64	$30 \to 1$	0.99987	0.00013	0.0047	+0.586	0.0007

10 Pre-registered predictions

Recorded in `PREREGISTRATION_MODELS.md` on 30 September 2026, before any run of this backbone.

Table 11: Pre-registered predictions and their outcomes.

	prediction	measured	outcome
P1	bank term equivalent to zero in each of 5 conditions	5 of 5	held
P2	static at width $2c$ beats `gap` at width c in each of 3 pairs	3 of 3	held
P3	routing term positive, $p < 0.05$, in at least 3 of 5 conditions	5 of 5	held

11 Complete results

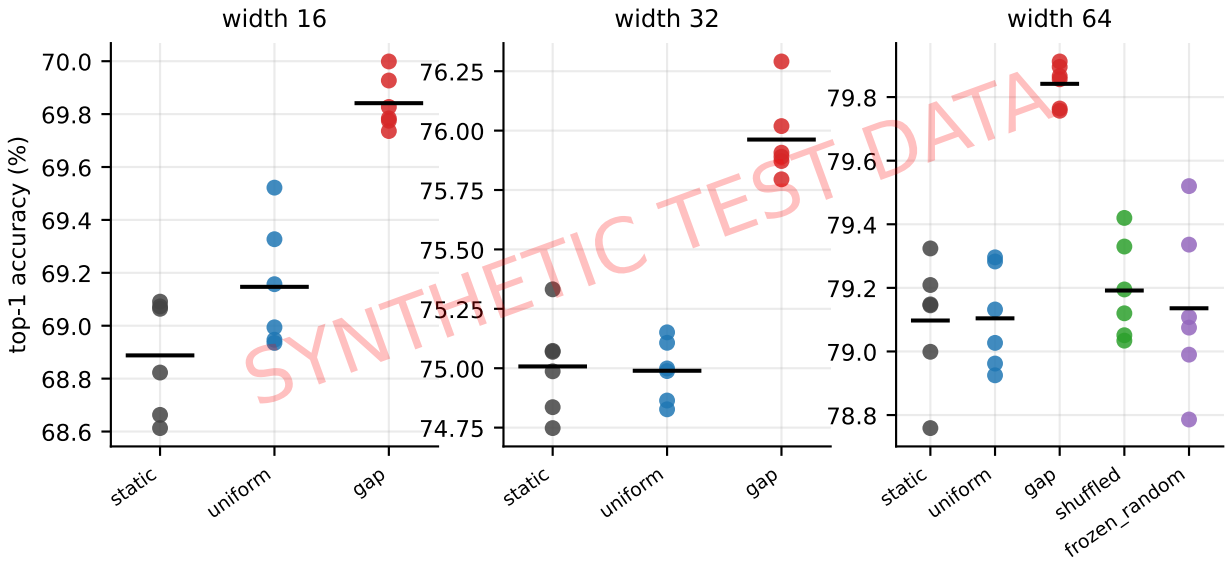


Figure 10: Every CIFAR-100 run at $K = 4$, $\tau = 1$. Bars are arm means.

Table 12: Every run at $K = 4$ and $\tau = 1$: final-epoch top-1 accuracy (%). – marks a missing or diverged run (Table 1).

dataset	width	arm	s0	s1	s2	s3	s4	s5	mean	sd
CIFAR-100	16	`static`	68.61	69.06	69.07	68.66	68.82	69.09	68.888	0.218
CIFAR-100	16	`uniform`	68.95	68.99	69.33	68.94	69.16	69.52	69.147	0.237
CIFAR-100	16	`gap`	69.83	69.78	69.93	69.74	70.00	69.78	69.841	0.101
CIFAR-100	32	`static`	74.75	75.07	75.33	74.84	75.07	74.99	75.007	0.205
CIFAR-100	32	`uniform`	75.11	74.86	75.15	74.83	75.00	74.99	74.989	0.128
CIFAR-100	32	`gap`	75.87	76.29	75.91	75.89	75.80	76.02	75.962	0.176
CIFAR-100	64	`static`	79.00	79.21	79.15	79.14	79.32	78.76	79.097	0.196
CIFAR-100	64	`uniform`	78.92	79.03	79.13	79.28	79.30	78.96	79.104	0.160
CIFAR-100	64	`gap`	79.89	79.87	79.76	79.91	79.76	79.86	79.842	0.066
CIFAR-100	64	`shuffled`	79.12	79.42	79.05	79.33	79.03	79.19	79.192	0.156
CIFAR-100	64	`frozen_random`	79.11	79.34	78.99	78.79	79.08	79.52	79.136	0.259
Tiny-ImageNet	32	`static`	60.05	59.79	59.73	60.21	59.94	60.25	59.991	0.215
Tiny-ImageNet	32	`uniform`	59.91	60.11	60.46	59.99	60.12	60.08	60.109	0.189
Tiny-ImageNet	32	`gap`	–	61.09	60.73	61.00	61.02	61.28	61.025	0.198
Tiny-ImageNet	64	`static`	65.07	64.86	64.66	65.17	65.23	65.29	65.046	0.243
Tiny-ImageNet	64	`uniform`	64.93	65.15	65.17	65.11	65.19	64.98	65.088	0.107
Tiny-ImageNet	64	`gap`	65.71	66.23	65.83	65.76	65.86	65.87	65.876	0.185

Table 13: Paired comparisons against `static`. p_W is the Wilcoxon signed-rank p (floor 0.031 at $n = 6$).

dataset	width	arm	Δ (pp)	95 % CI	p_t	p_W	d_z	p_{TOST}	verdict
CIFAR-100	16	`uniform`	+0.259	[+0.078, +0.440]	0.0144	0.0625	+1.50	0.0095	higher
CIFAR-100	16	`gap`	+0.954	[+0.710, +1.197]	0.0002	0.0312	+4.12	0.9975	higher
CIFAR-100	32	`uniform`	-0.018	[-0.232, +0.195]	0.8356	0.5625	-0.09	0.0011	equivalent
CIFAR-100	32	`gap`	+0.955	[+0.692, +1.218]	0.0002	0.0312	+3.81	0.9966	higher
CIFAR-100	64	`uniform`	+0.007	[-0.141, +0.155]	0.9102	1.0000	+0.05	0.0002	equivalent
CIFAR-100	64	`gap`	+0.744	[+0.501, +0.988]	0.0005	0.0312	+3.21	0.9753	higher
CIFAR-100	64	`shuffled`	+0.094	[-0.173, +0.361]	0.4054	0.4375	+0.37	0.0057	equivalent
CIFAR-100	64	`frozen_random`	+0.039	[-0.385, +0.462]	0.8245	0.8438	+0.10	0.0190	equivalent
Tiny-ImageNet	32	`uniform`	+0.118	[-0.272, +0.508]	0.4723	0.5625	+0.32	0.0266	equivalent
Tiny-ImageNet	32	`gap`	+1.044	[+0.816, +1.273]	0.0002	0.0625	+5.68	0.9987	higher (n=5)
Tiny-ImageNet	64	`uniform`	+0.042	[-0.276, +0.359]	0.7498	1.0000	+0.14	0.0069	equivalent
Tiny-ImageNet	64	`gap`	+0.830	[+0.464, +1.197]	0.0021	0.0312	+2.38	0.9659	higher

Table 14: Configuration and cost. Router parameters in thousands; training loss is the final-epoch mean over seeds; minutes are logged wall clock.

dataset	width	arm	params (M)	router (K)	train loss	min/run
CIFAR-100	16	`static`	1.53	0.0	0.1119	10.0
CIFAR-100	16	`uniform`	3.65	0.0	0.1119	40.0
CIFAR-100	16	`gap`	3.68	22.8	0.0819	40.0
CIFAR-100	32	`static`	5.99	0.0	0.0619	20.0
CIFAR-100	32	`uniform`	14.48	0.0	0.0619	80.0
CIFAR-100	32	`gap`	14.56	81.9	0.0469	80.0
CIFAR-100	64	`static`	23.71	0.0	0.0244	40.0
CIFAR-100	64	`uniform`	57.66	0.0	0.0244	160.0
CIFAR-100	64	`gap`	57.98	319.2	0.0206	160.0
CIFAR-100	64	`shuffled`	57.98	319.2	0.0244	160.0
CIFAR-100	64	`frozen_random`	57.66	0.0	0.0244	160.0
Tiny-ImageNet	32	`static`	6.09	0.0	0.0869	20.0
Tiny-ImageNet	32	`uniform`	14.58	0.0	0.0869	80.0
Tiny-ImageNet	32	`gap`	14.66	81.9	0.0644	80.0
Tiny-ImageNet	64	`static`	23.91	0.0	0.0306	40.0
Tiny-ImageNet	64	`uniform`	57.86	0.0	0.0306	160.0
Tiny-ImageNet	64	`gap`	58.18	319.2	0.0250	160.0

Table 15: Every run of this backbone. GPU-hours from logged wall clock.

dataset	width	K	τ	arms	runs	GPU-h
CIFAR-100	16	4	constant	`static`, `uniform`, `gap`	18	9.0
CIFAR-100	32	4	constant	`static`, `uniform`, `gap`	18	18.0
CIFAR-100	64	4	constant	`static`, `uniform`, `gap`, `shuffled`, `frozen_random`	30	68.0
CIFAR-100	32	2	constant	`gap`	6	8.0
CIFAR-100	32	8	constant	`gap`	6	8.0
CIFAR-100	16	4	annealed	`gap`	6	4.0
CIFAR-100	32	4	annealed	`gap`	5	6.7
CIFAR-100	64	4	annealed	`gap`	6	16.0
Tiny-ImageNet	32	4	constant	`static`, `uniform`, `gap`	17	16.7
Tiny-ImageNet	64	4	constant	`static`, `uniform`, `gap`	18	36.0
total					130	190.3